Improving Pupil Detection on a Robotically Aligned OCT System with Patterned LED Illumination

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OCT INTRODUCTION

Optical Coherence Tomography (OCT) is an important imaging technique used widely by ophthalmologists to diagnose and manage common eye diseases.

Limitation to Clinical OCT System:
- Limited to tabletop devices
- Requires the patient to be upright and still while scanning
- Excludes unconscious, bedbound, or injured patients and those with movement and cervical mobility disorders

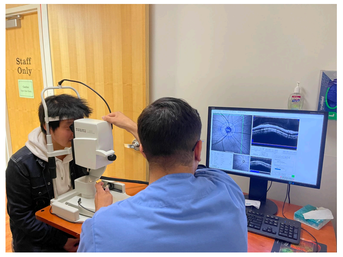


Figure 1: Clinical OCT

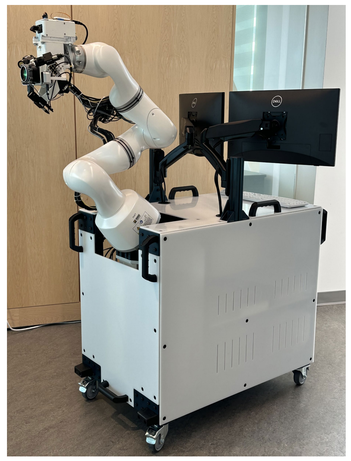


Figure 2: Mobile RAOCT system with a wheeled cart housing

Robot OCT System:
- Unconstrained patient posture
- Robot autonomous alignment (minimum user skill requirement and patient cooperation)
- Wheeled cart for mobility

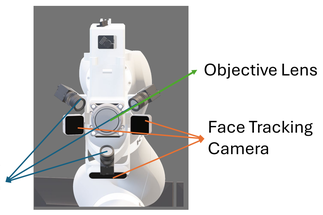


Figure 3: Camera and lens setup of the robot OCT system [1]

MOTIVATION

Importance of Reliable Pupil Tracking:
Accurate alignment of the pupil with the OCT scanning laser is essential for acquiring high-quality scans and reliable retinal measurements. In robotically aligned OCT systems, robust and real-time pupil tracking becomes even more critical, particularly when imaging patients with movement disorders such as Parkinson's disease.

Current Pupil Tracking System:
We currently use a 4-LED system with 3 cameras. The LEDs are 850nm, which allows for reflections in the cornea. The camera then detects the reflection and uses it to determine the pupil position.

Current Pupil Tracking Limitations
- Low robustness
- Poor integration with the rest of the OCT system
- Inconsistent reflection detection

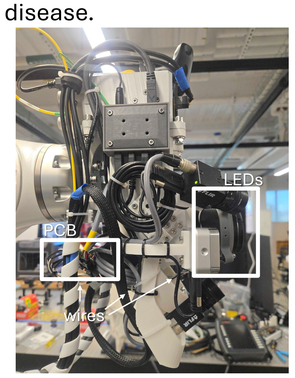


Figure 4: Image of current system

PCB DESIGN

To address previous limitations, we developed a custom, ring-shaped PCB in Altium, equipped with eight LEDs that pulse in synchronized patterns.

Design Specifications:
- Utilizes Blackfly S standard wiring for synchronized camera capture
- Fits on the lens with 10 mm in width
- Eight 900 nm LED with a switching system by trigger

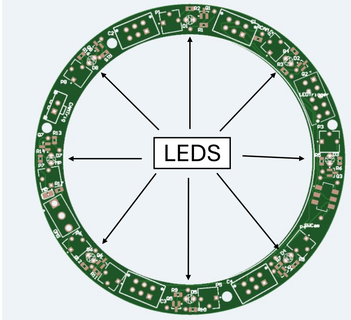


Figure 5: PCB footprint of the board

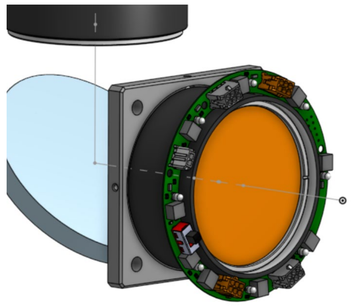


Figure 6: Board fitted on the outside of the lens in the CAD

Benefits of the Improvements
- An increased number of LED allow for pulsing in synchronized patterns
- Ring shape allows for tight integration around the lens, optimizing packaging.
- Use standard connectors to improve connection mobility

THEORY OF OPERATION

To track the pupil, we use eight LEDs. We alternate by flashing four LEDs at a time while capturing images with the camera. By analyzing the positions of the reflections of the LED and comparing them with the previous frame, we can accurately identify the LED reflection location and calculate the pupil's location.

Pupil Tracking Calculation
The system determines the pupil's position by analyzing the geometric relationships between the detected LED reflections and the expected reflection points. Solving for c in Figure 8 gets the pupil location.

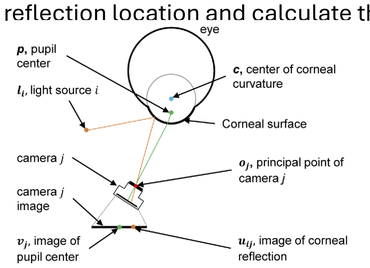


Figure 7: Variables in the equation below [2]

$$\underbrace{\begin{bmatrix} [(\mathbf{l}_1-\mathbf{o}_1)\times(\mathbf{u}_{11}-\mathbf{o}_1)]^T \\ [(\mathbf{l}_1-\mathbf{o}_2)\times(\mathbf{u}_{12}-\mathbf{o}_2)]^T \\ \vdots \\ [(\mathbf{l}_N-\mathbf{o}_1)\times(\mathbf{u}_{N1}-\mathbf{o}_1)]^T \\ [(\mathbf{l}_N-\mathbf{o}_2)\times(\mathbf{u}_{N2}-\mathbf{o}_2)]^T \end{bmatrix}}_{\mathbf{M}_2} \mathbf{c} = \underbrace{\begin{bmatrix} [(\mathbf{l}_1-\mathbf{o}_1)\times(\mathbf{u}_{11}-\mathbf{o}_1)]\bullet\mathbf{o}_1 \\ [(\mathbf{l}_1-\mathbf{o}_2)\times(\mathbf{u}_{12}-\mathbf{o}_2)]\bullet\mathbf{o}_2 \\ \vdots \\ [(\mathbf{l}_N-\mathbf{o}_1)\times(\mathbf{u}_{N1}-\mathbf{o}_1)]\bullet\mathbf{o}_1 \\ [(\mathbf{l}_N-\mathbf{o}_2)\times(\mathbf{u}_{N2}-\mathbf{o}_2)]\bullet\mathbf{o}_2 \end{bmatrix}}_{\mathbf{h}}$$

Figure 8: Equation for the pupil calculation c [3]

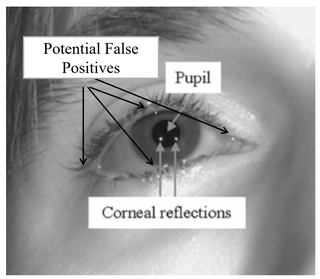


Figure 9: Examples of the potential false positives [3]

Benefits of the New Method
- Sequential LED flashing facilitates easy identification of LED reflections and reduces the impact of noise
- Decreases calculation time for corneal reflection segmentation

NEXT STEPS

To test the reliability of this new system, we characterize 3 aspects:
- The accuracy of range of the detection
- The accuracy of speed of the detection
- How it works with different irises

Setup
To test the system, we will place a model eye on a motorized linear stage 70 mm in front of the scanner. Put on a mount to allow for rotary movement

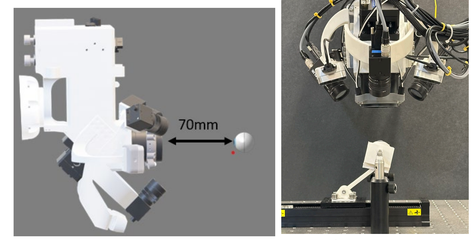


Figure 10: Model of the setup [4]

Testing for Range
We measure the range of motion for the eye and accuracy at various positions
1. Move the eye model across the motorized stage at known increments (1 mm)
2. Compare values with the expected location known by the motorized stage
3. Compute errors for position

Testing for Speed
We want to test the accuracy and limits of motion for the eye as the movement speed increases
1. Move the eye model across the motorized stage at known increments (1 mm)
2. Gradually increase the speed of the motorized stage
3. Compute errors from known positions

Testing on Irises
Different iris colors affect detection of NIR (Near Infrared) LED reflections. To test the improvements of changing the LED wavelength, we will test the tracking on a series of people with different irises.

CONCLUSION

- Accurate pupil tracking is essential for reliable B-scan acquisition in OCT imaging.
- Enhancements include an increased number of LEDs for better tracking performance and optimized packaging for improved integration and accessibility.
- The new pupil tracking method promises to improve identification of NIR reflections, reduce noise, and decrease calculation time.
- Next steps involve testing the system's accuracy with range and speed, as well as robustness with a variety of irises

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[1] G. Zhou et al., "Mobile Robotic Optical Coherence Tomography System for Ophthalmic Imaging in Clinical Environments," in 2025 International Symposium on Medical Robotics (ISMR), Atlanta, GA, USA: IEEE, May 2025, pp. 1–7. doi: 10.1109/ISMR67322.2025.11025963.
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